Module 0 Final Exam

Provide the best answer for each of the following questions.

Time Remaining 01:14:17

## Main Question Set

Solve six of the first ten problems to get a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every single one of them!

## 1

Review Later

Which of these numbers cannot be written as $n^4$, where $n$ is an integer?

- 228886641
- 60037250625
- 12745506813

## 2

Review Later

What is the simplified value of

$$\frac{5}{6} - \frac{6}{7} + \frac{1}{21}?$$

Your answer should be a fraction without spaces.

For example, if the answer were $\frac{3}{2}$, you would enter $3$ in the first blank, and $2$ in the second blank.

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## 3

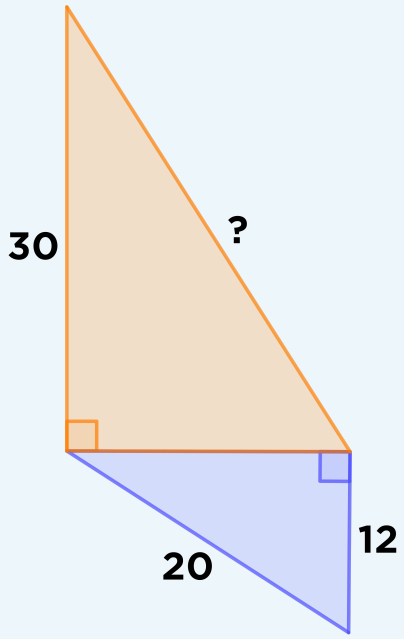
Review Later

Every two hours, a sardine escapes out of a fishing net and travels downstream at $3$ miles per hour. How many miles are there between each of these sardines?

## 4

Review Later

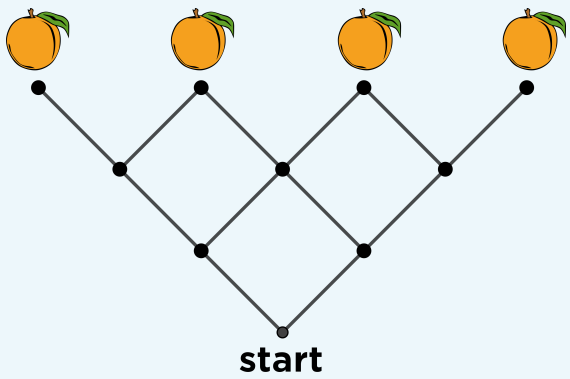
What is the missing side length, marked with a "?", in the triangle below?




## 5

Review Later

How many ways are there to start from the bottom of this tree and climb up to get any of the fruits on the top, if you must always go up with every step?




## 6

Review Later

Five rectangles, $A, B, C, D$, and $E$ have integer side lengths. Rectangle $A$ has a width of $15$, rectangle $B$ has a width of $35$, rectangle $C$ has a length of $21$, rectangle $D$ has a length of $45$, and rectangle $E$ has a length of $63$.

If all five rectangles have the same area $K$, what is the least possible value of $K$?

## 7

Review Later

What is the biggest $2$-digit number that has an odd number of factors?

## 8

Review Later

How many factors of the number $784$ are perfect squares?

## 9

Review Later

How many $3$-digit numbers can be made out of only even digits? (Zero counts as an even digit.)

## 10

Review Later

If $a$, $b$, and $c$ are numbers that satisfy:

$$a^2b = 2$$
$$b^2c = 28$$
$$c^2a = 49,$$

then what is the value of $(abc)^2$?

## Bonus

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

## 11

Review Later

At a school, every student likes either cats or dogs (but not both), and likes either apples or bananas (but not both). Also, $\frac{1}{3}$ of the students who like cats also like apples, and $\frac{1}{3}$ of the students who like apples also like dogs. Lastly, $\frac{3}{4}$ of the students who like dogs also like bananas.

What percent of the students like apples?

Your answer should be the number only, without the % sign.

## 12

Review Later

How many ways are there for $13$ nurses to wear either pink or blue gloves if there are one dozen of each color?

## 13

Review Later

A scientist is studying a cell under a microscope. First, she uses a lens that magnifies $200$x (that makes the length and width dimensions of the cell look $200$ times longer), but the cell is still too small. If the scientist uses a $700$x lens instead, how many times as much area will the cell appear to take up compared to when she uses the $200$x lens?

Your answer should be a decimal rounded to the nearest hundredth without spaces, e.g. 1.21

## 14

Review Later

There are $10$ empty parking spaces in a row at a car dealership lot. Three identical cars drive into the parking lot, each of them choosing a random empty parking space to park in. Since the cars are new, they don't have license plates.

What is the probability that all $3$ cars are parked next to each other?

Your answer should be a fraction in simplest form with no spaces.

For example, if the answer were $\frac{3}{2}$, you would enter $3$ into the first blank and $2$ into the second blank.

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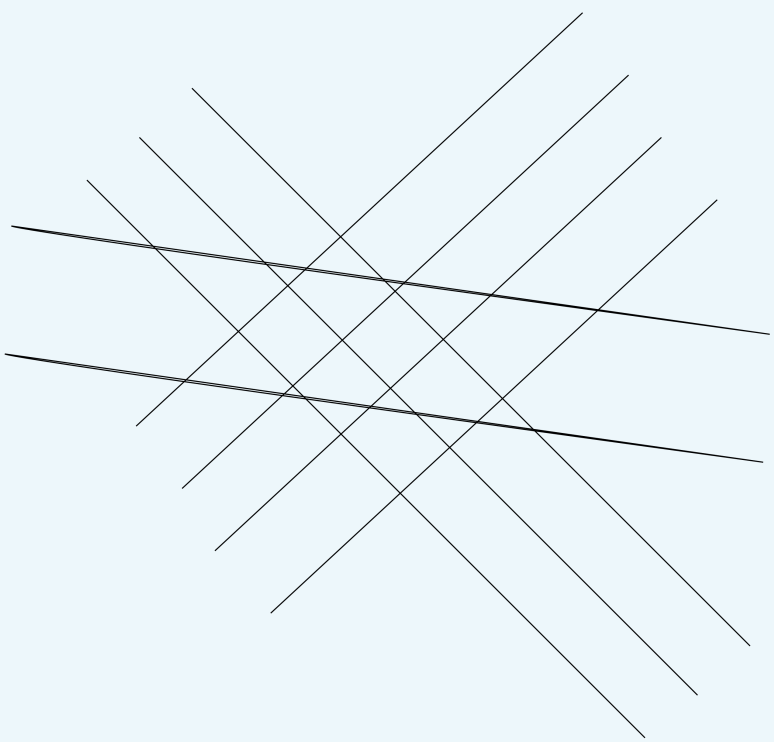
## 15

Review Later

How many triangles are in this figure?

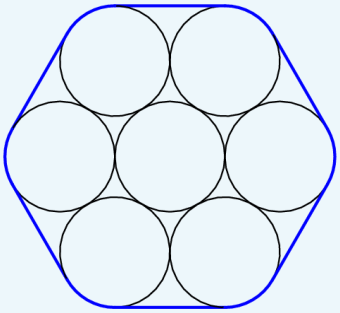


## 16

Review Later

As shown below, $7$ circular pegs of the same radius are bundled together so that they are tangent to one another, and a string is wrapped around the outside. If the pegs' radii are all increased by $2$ inches, how much longer would the string around the pegs be, in inches?

Your answer should be a decimal rounded to the nearest hundredth.



## 17

Review Later

Of the people tested for a virus, $12\%$ of them were incorrectly tested to be positive, $8\%$ were incorrectly tested to be negative, and $48\%$ of the positive test results are incorrect. What percent of the people tested have the virus?

Your answer should be the number only, without units.

## 18

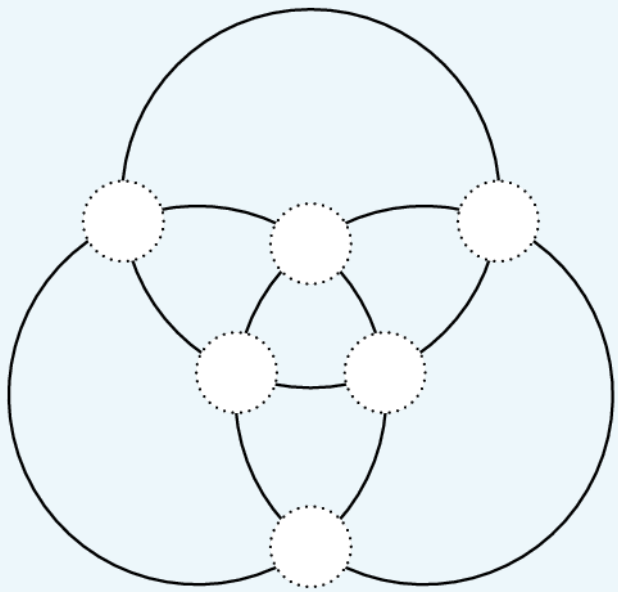
Review Later

How many regular polygons have an interior angle that is an integer number of degrees in measure?

## 19

Review Later

Three circles are drawn to intersect each other like a Venn Diagram, and there is a blank spot on each of the $7$ intersection points, as shown.



Each blank spot is filled with one of these six numbers: $10, 11, 12, 13, 14,$ and $15$. The sum of the numbers on each circle is the same. What is the sum of the numbers on one circle?

## 20

Review Later

A salmon starts swimming upstream at $6$ miles per hour, immediately encountering one of many sardines swimming downstream at $3$ miles her hour. The sardines are released from the beginning of the stream at the rate of one every $2$ hours. If the salmon swims the length of the $60$ mile stream, assuming it eats every sardine it encounters, how many sardines will it eat?

View Marked Questions

Finished

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